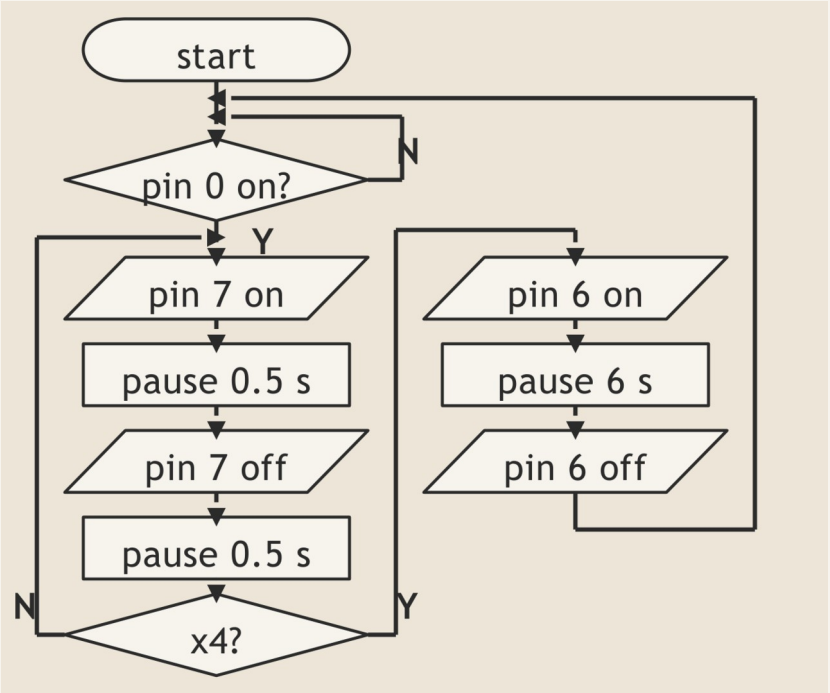
Daily Review



1	A microcontroller runs a stored…	Program
2	One advantage in the MI's words?	It can be reprogrammed
3	An input switch goes from … to the pin.	5 V
4	A motor needs what between it and a pin?	A transistor
5	'Driver' on its own scores…	0 marks

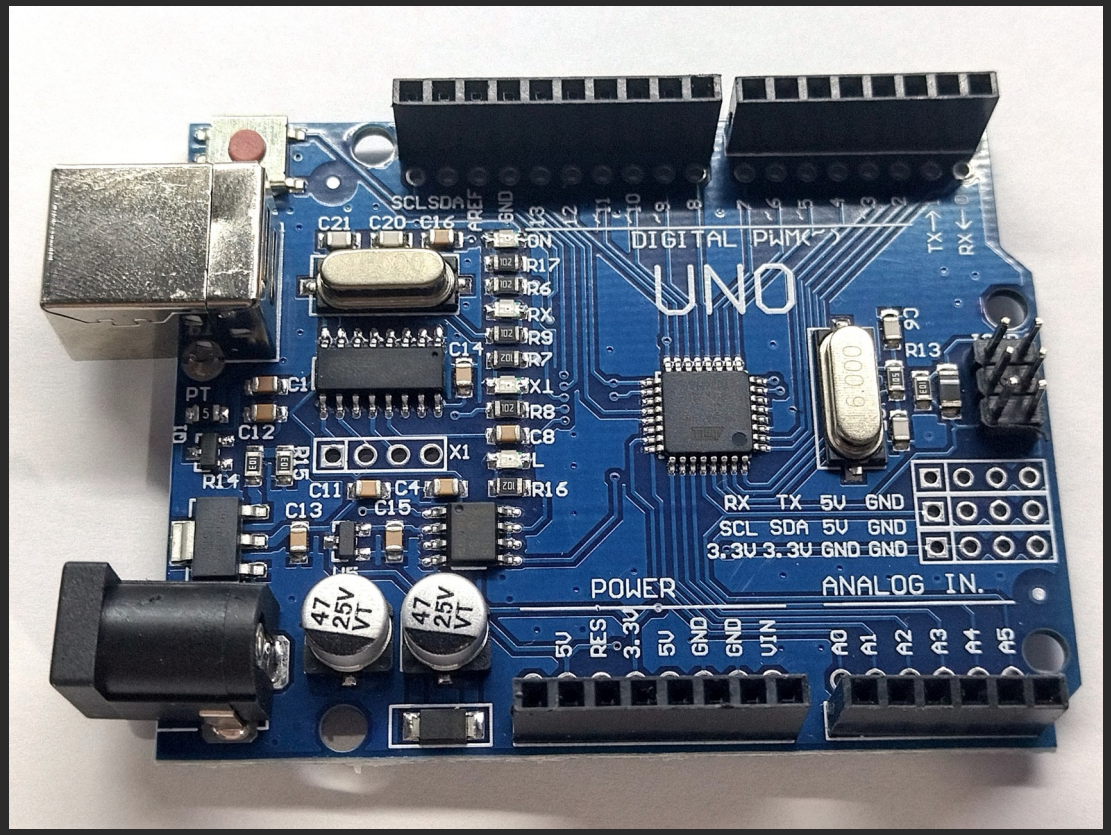
Flowcharts

How do you tell a chip what to do, and when?



National 5 Engineering Science

How do you tell a chip: flash the lamp 3 times, then wait for the button?



The Uno we use

PART 1

Symbols and Structures

Six symbols. Four ways to put them together.

By the end of this part…



1. I can identify the six flowchart symbols
2. I can describe a continuous loop and a fixed loop
3. I can describe a time delay and a branch

What a Flowchart Is

A flowchart

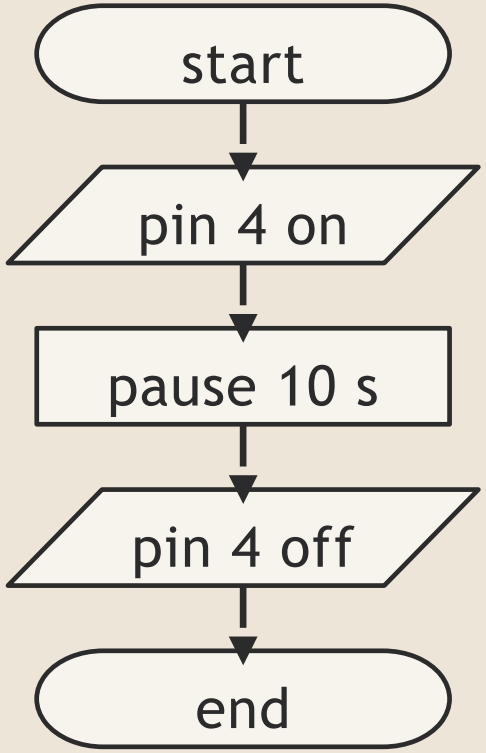
the steps of a program, in order

Each step

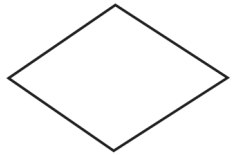
one box

The lines

show the order



The Six Symbols

Terminator symbol		Used for the start and end of a main program or sub-procedure.
Line symbol		Shows the direction of program flow. For flow down or to the right, an arrow is not needed. For flow upwards or to the left, arrows are added.
Input/Output		Used to control outputs or to show that data is being received.
Process symbol		Used for operations which take place within the microcontroller, for example a delay.
Decision symbol		Program flow is determined by a “yes” or “no” answer to the question in the box.
Sub procedure symbol		Used to call a sub-procedure.

Lines and Arrows

Down or right

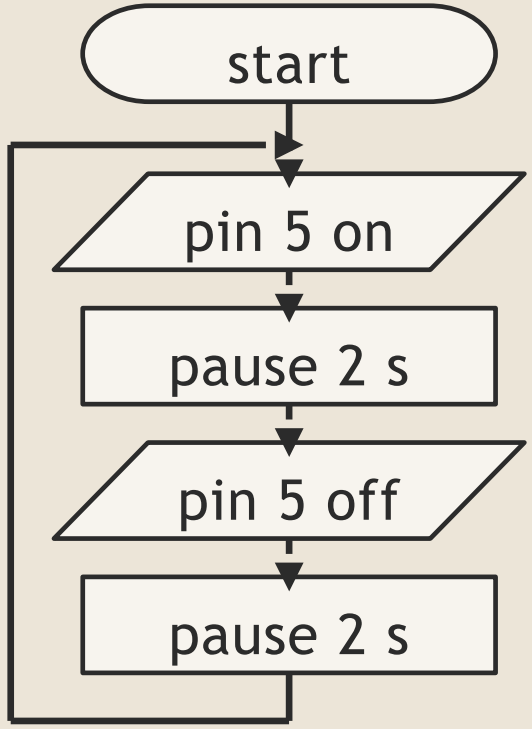
an arrow is not needed

Up or left

an arrow is needed

Every loop back

draw its arrow: the MI marks it



What Goes in the Box



Output

pin 7 on · pin 7 off

Input decision

pin 1 on?

Delay

pause 0.5 s (always a unit)

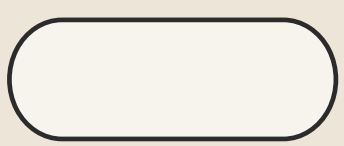
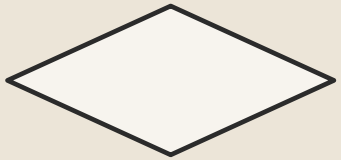
Fixed loop

x3?

The booklet writes

Lamp (output, pin 13) ON
Wait 1 s (1000 ms)
count < 5?

Name the Symbol

1	2	3	4	5
				
Terminator	Decision	Input or output	Process	Sub procedure

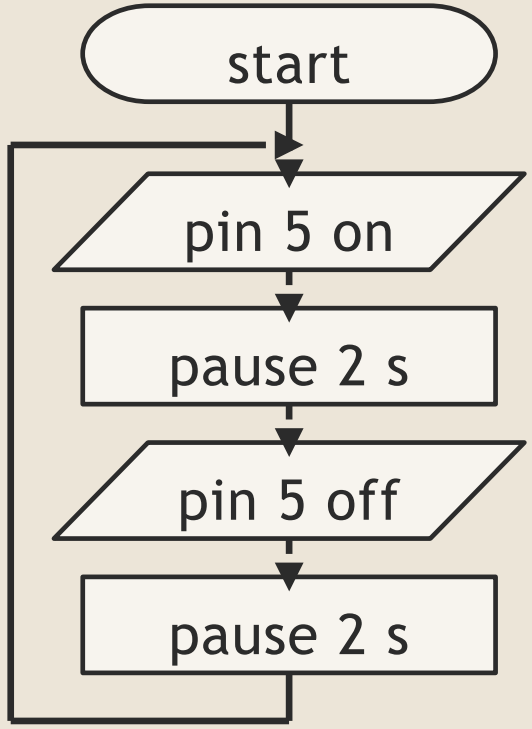
Continuous Loop

Repeats

forever, until the power goes off

Drawn as

a line back up, with an arrow



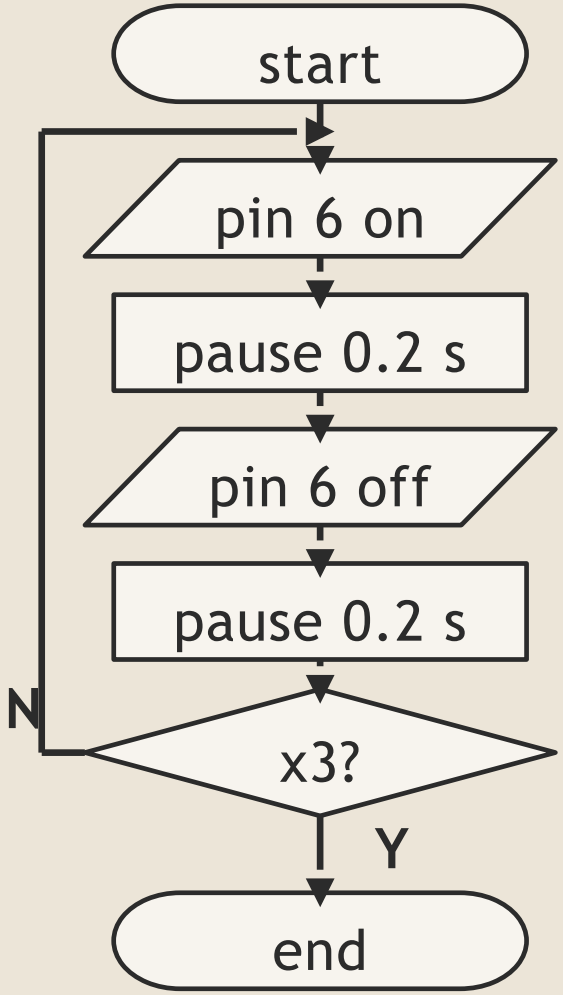
Fixed Loop

Repeats

a set number of times, then moves on

Drawn as

a decision, x3?, N loops back



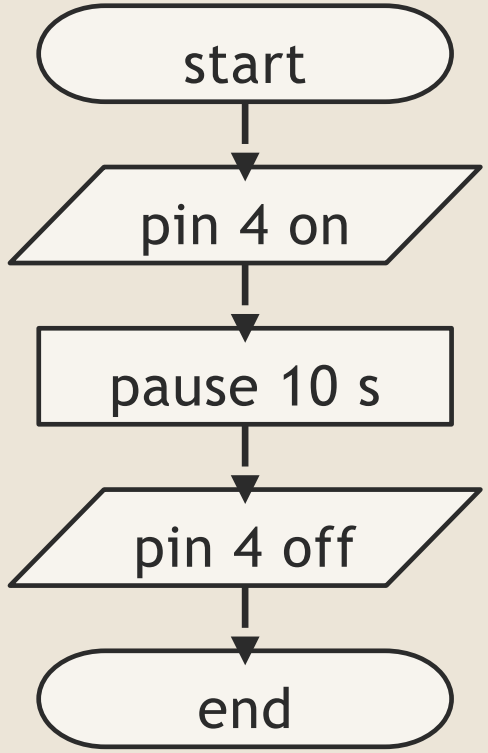
Time Delay

A pause

the program waits for a set time

Meanwhile

outputs stay as they are



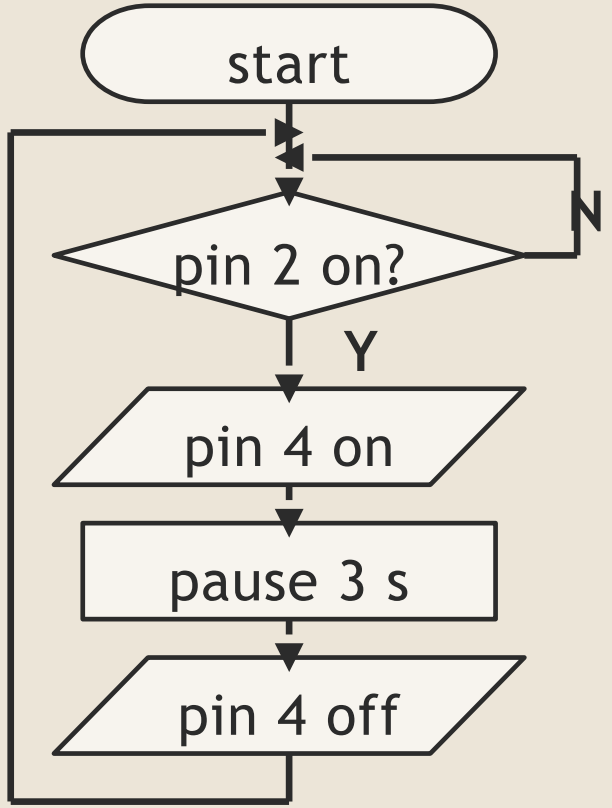
Branch

A decision

yes or no

Each answer

its own path, labelled Y and N



Which Structure?



1. A pump runs for 5 minutes, then stops.
2. A sign flashes all day.
3. A bell rings 4 times, then stops.
4. If the door opens, sound the alarm.
5. A fan waits for the switch, every time.

Time delay

Continuous loop

Fixed loop

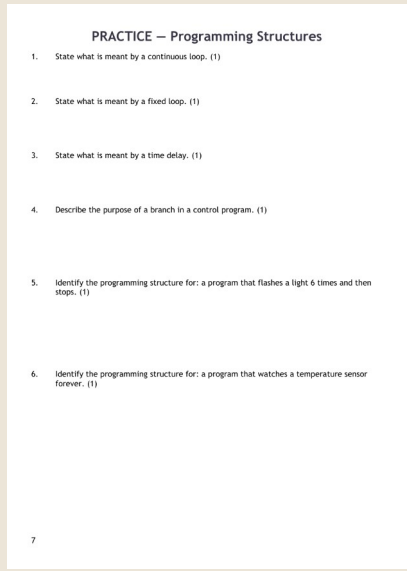
Branch

Branch, in a continuous loop

PRACTICE: Programming Structures

Booklet page 7

Answer Q1 to Q6 on your own.
One sentence each.



PRACTICE — Programming Structures

1. State what is meant by a continuous loop. (1)
2. State what is meant by a fixed loop. (1)
3. State what is meant by a time delay. (1)
4. Describe the purpose of a branch in a control program. (1)
5. Identify the programming structure for: a program that flashes a light 6 times and then stops. (1)
6. Identify the programming structure for: a program that watches a temperature sensor forever. (1)

7

6 min

Programming Structures: Answers



1	Q1 Continuous loop	Repeats forever, until power is removed
2	Q2 Fixed loop	Repeats a set number of times
3	Q3 Time delay	A set pause; outputs stay the same

Programming Structures: Answers



4	Q4 Branch
5	Q5 and Q6

A yes/no decision picks one of two paths

Q5 fixed loop; Q6 continuous loop

Checkpoint



1. Which symbol holds a delay?

   Process (rectangle)

2. Which line needs an arrow?

   Any line going up or left

3. Why does a control program loop forever?

   Watches inputs — **so it reacts any time**

Plenary: Part 1



1	The diamond is the … symbol.	Decision
2	Input and output share which shape?	Parallelogram
3	A loop that repeats forever is…	A continuous loop
4	x3? makes what kind of loop?	A fixed loop
5	During a delay, the outputs…	Stay as they are

Look back — by the end of this part…



1. I can identify the six flowchart symbols
2. I can describe a continuous loop and a fixed loop
3. I can describe a time delay and a branch

Daily Review



1	The diamond is the … symbol.	Decision
2	A delay goes in which box?	Process (rectangle)
3	A fixed loop repeats…	A set number of times
4	Arrows are needed going…	Up or left
5	A decision needs which two labels?	Y and N

PART 2

The Exam Flowchart

10 marks, every year. Here is how they are given.

By the end of this part…



1. I can complete a flowchart from an I/O table
2. I can put every pin number and unit in
3. I can draw every decision with Y, N, a loop and an arrow

How the Marks Are Given



Each output: on and off, in the right place (1)

Each input decision: Y, N, a loop, an arrow (1)

Each delay: right place, with a unit (1)

x3?: Y, N, and an arrow back before the repeat (1)

A loop to the start with an arrow, or an end (1)

All the symbols right (1)

The Method



1 **Read** the I/O table: which pins are in, which are out.

2 **Draw** one box per bullet, in order, top to bottom.

3 **Turn** each input into a decision, pin 1 on?, with Y and N.

4 **Turn** each repeat into x3?, with an arrow back.

5 **Finish:** loop to start, or end. Check pins, units, arrows.

Worked Example: The Question (9 marks)



Input

train sensor: pin 0

Outputs

lamps: pin 7

barrier motor: pin 6

When the train sensor is on:

the lamps flash on and off 4 times, 0.5 s on and 0.5 s off;

the barrier motor then turns on for 6 s, then off;

the sequence returns to the start.

Worked Example: The Answer (9 marks)

pin 0 on?: Y, N, loop (1)

pin 7 on, off (1)

Both pauses, units (1)

x4?: Y, N (1)

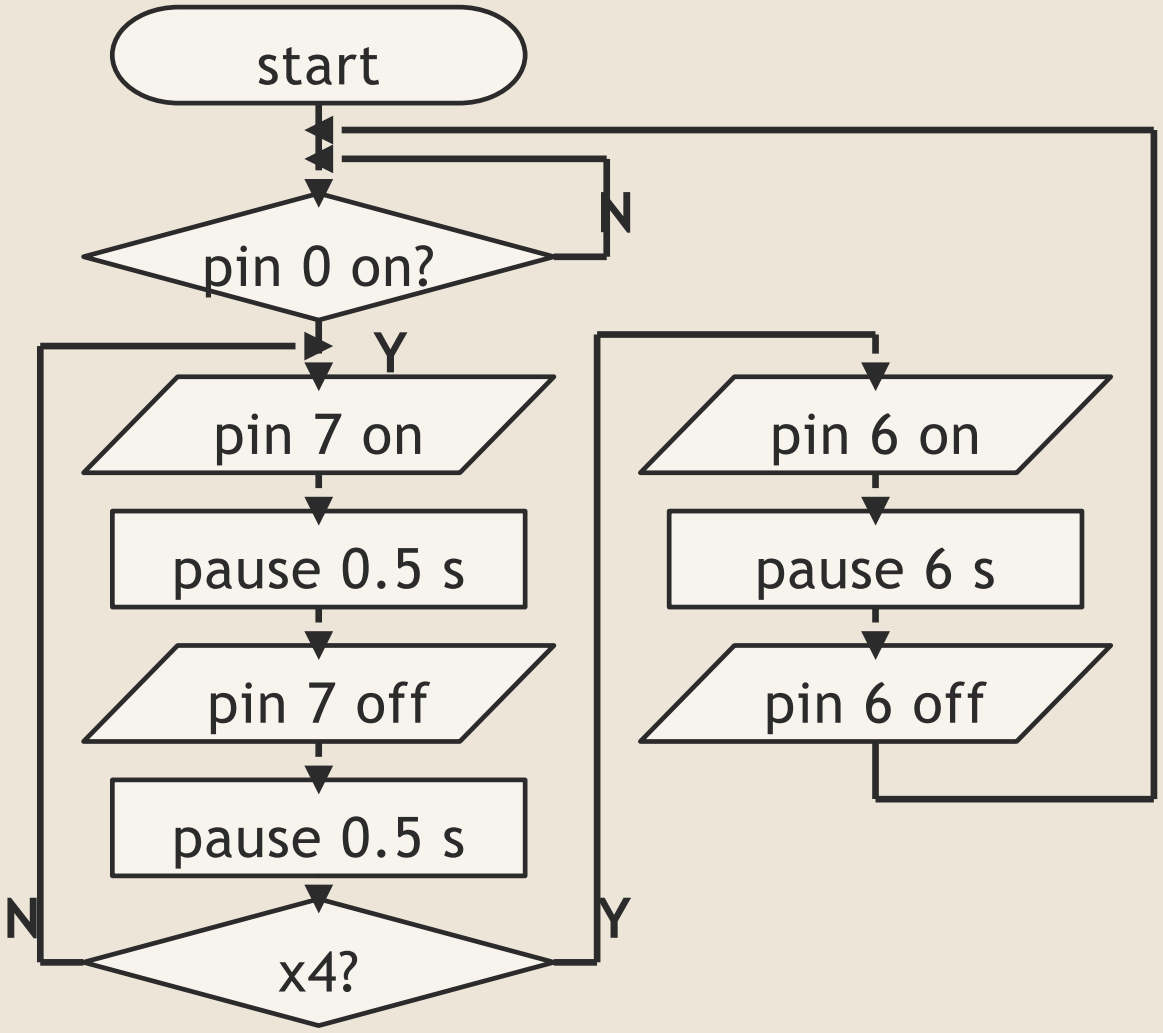
Arrow back (1)

pin 6 on, off (1)

pause 6 s, unit (1)

Loop to start (1)

All symbols right (1)



Your Turn: The Car Wash (6 marks)

Input

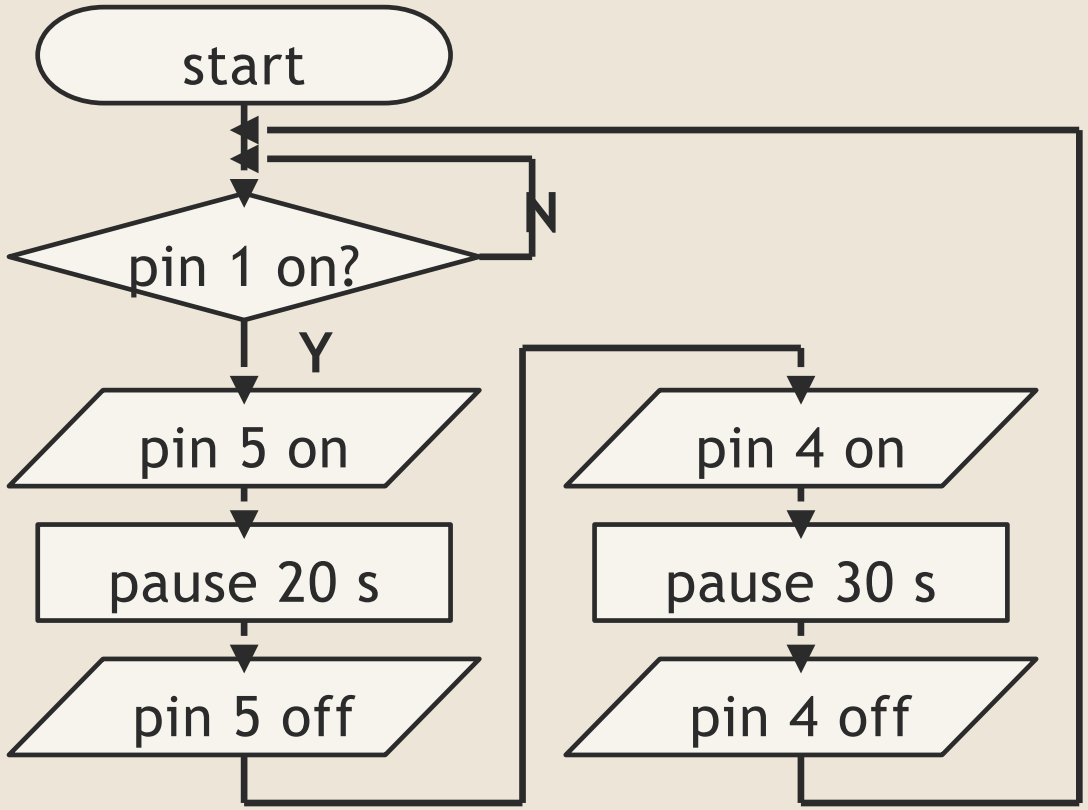
button: pin 1

Outputs

pump: pin 5; brush: pin 4

Pressed

pump on 20 s, off; brush on 30 s, off; repeat



Spot the Mistake

A pupil drew this

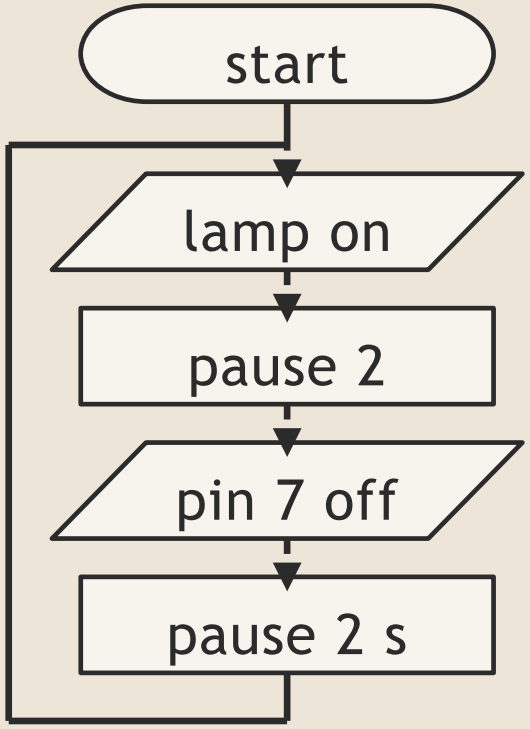
a lamp flashing for ever

Find

three mistakes

Each one

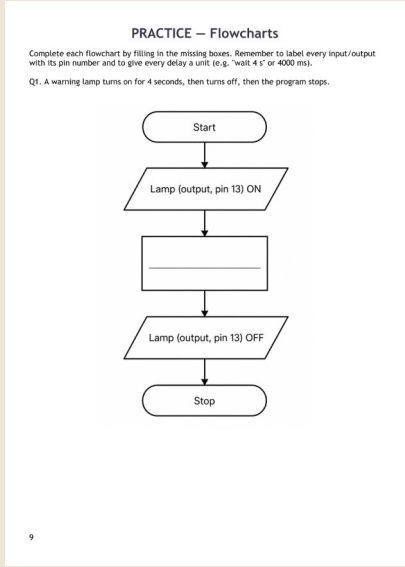
loses a mark



PRACTICE: Flowcharts

Booklet pages 9-13

Q1 to Q5: fill in the missing boxes.
Every pin number. Every unit.


PRACTICE — Flowcharts

Complete each flowchart by filling in the missing boxes. Remember to label every input/output with its pin number and to give every delay a unit (e.g. "wait 4 s" or 4000 ms).

Q1. A warning lamp turns on for 4 seconds, then turns off, then the program stops.

Start

Lamp (output, pin 13) ON

Lamp (output, pin 13) OFF

Stop

9

12 min

Flowcharts: Answers

1. Q1 the blank box

2. Q2 the three blanks

3. Q3 the four blanks

Wait 4 s (4000 ms)

Wait 5 s; wait 2 s; buzzer pin 8 off

LED on; wait 1 s; LED off; wait 1 s

Flowcharts: Answers



4 Q4 the three blanks

Pressed?; buzzer on 2 s; buzzer off

5 Q5 the decision

count < 5? (Yes loops back)

Checkpoint



1. What must every output box show?

   Its pin number

2. What must every delay show?

   Its unit

3. Why draw the arrow on a loop back?

   It goes up — **so the MI needs the arrow**

Plenary: Part 2



1	An input becomes which box?	A decision: pin 1 on?
2	A repeat becomes…	x3? with an arrow back
3	The N loop of x3? goes back to…	Before the repeated part
4	Extra boxes in the exam are…	Ignored
5	The program ends with…	A loop to start, or end

Look back — by the end of this part…



1. I can complete a flowchart from an I/O table
2. I can put every pin number and unit in
3. I can draw every decision with Y, N, a loop and an arrow

Daily Review



1	Every output box shows its…	Its pin number
2	Every delay shows its…	Unit
3	A loop back needs an…	Arrow
4	x3? is what kind of loop?	Fixed
5	Extra steps in the exam are…	Ignored

PART 3

Build It, Test It

Draw it, then make it run.

By the end of this part…



1. I can build a flowchart in NoSyntaxSim and test it
2. I can build the same flowchart in Yenka
3. I can answer a past-paper flowchart question

NoSyntaxSim: Flowchart to Uno

Each box

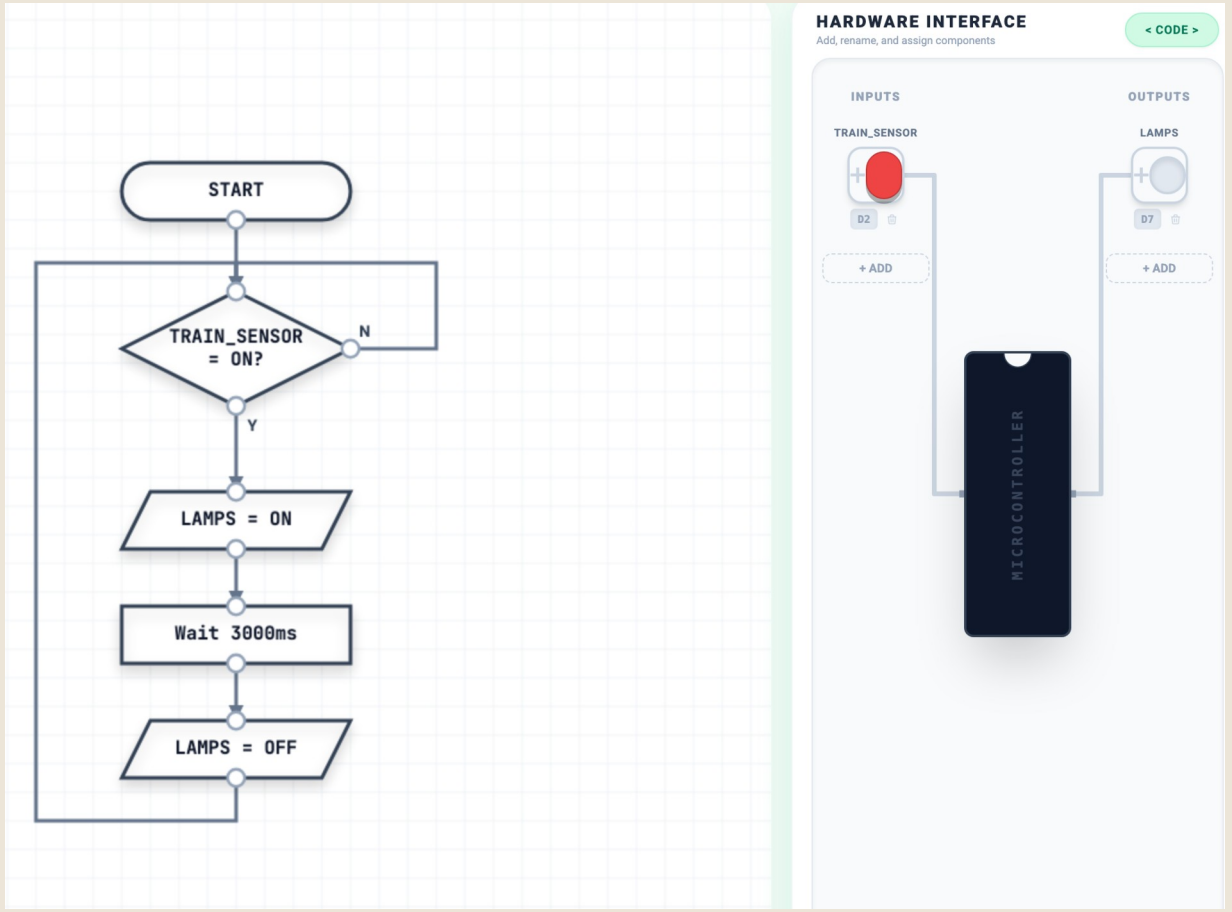
picks its pin

Run Programme

lights each step

On paper

still pin 0 on?, pause 3 s



Yenka: The Same Flowchart

Same symbols

same order

Each output

set its pin

Each wait

set its time and unit

Run it

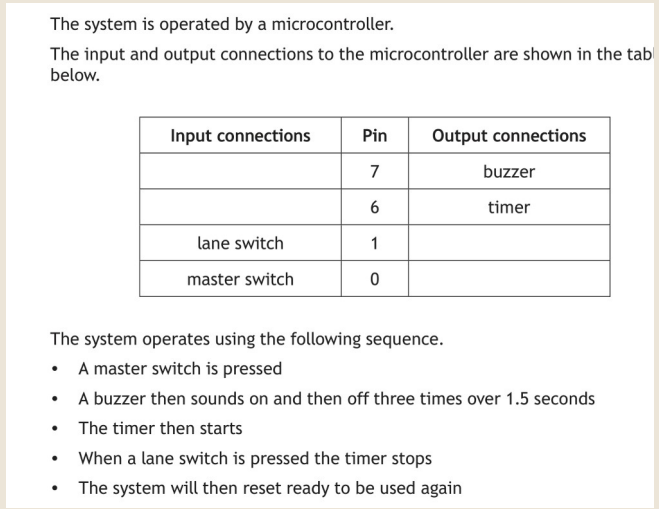
press the input, watch the output

Build It: 2022 Q9 in Both Simulators

NoSyntaxSim, then Yenka

On paper: use the question's pins.

NoSyntaxSim: pin 0 on D2, pin 1 on D3.



The system is operated by a microcontroller.

The input and output connections to the microcontroller are shown in the tab
below.

Input connections	Pin	Output connections
	7	buzzer
	6	timer
lane switch	1	
master switch	0	

The system operates using the following sequence.

- A master switch is pressed
- A buzzer then sounds on and then off three times over 1.5 seconds
- The timer then starts
- When a lane switch is pressed the timer stops
- The system will then reset ready to be used again

15 min

Past Papers: Flowcharts

The dishwasher is operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table be

Input connection	Pin	Output connection
	7	red LED
	6	water pump
	5	heating element
	4	buzzer
	3	
start switch	2	
door sensor	1	
	0	

A drying cycle used in the dishwasher operates using the following sequence:

- When a start switch is pressed a red LED will turn on.
- After a door sensor is activated a water pump will turn on for 35 seconds a then turn off.
- A heating element will then turn on for 5 minutes and then turn off.
- A buzzer will then sound six times, turning on for 0.3 seconds and turning for 0.3 seconds each time.
- The red LED will then turn off.
- The sequence will return to the start.

2024 Q11(b) (11 marks)

In class

2022 Q9(a), 2024 Q11(b)

Homework

2021 Q13(b), 2023 Q8(a)

2022 Q9(a): 10 Marks, 1 per Line



1 pin 0 on?: Y/N, loop, arrow

2 pin 7 on, off in place

3 Both delays in place

4 0.5 s per on-off, unit

5 x3?: Y and N

6 Arrow back before pin 7 on

7 pin 6 on, off in place

8 pin 1 on?: Y/N, loop, arrow

9 Loop to start, arrow

10 All symbols right

2024 Q11(b): 11 Marks, 1 per Line



1 pin 7 on, off in place

2 pin 1 on?: Y and N

3 pin 1 loop, arrow, before it

4 pin 6 on, off in place

5 pin 5 on, off in place

6 pin 4 on, off in place

7 All 4 delays, units

8 x6?: Y and N

9 Arrow back before pin 4 on

10 Loop to start, arrow

11 All symbols right

Plenary: Part 3

1	In NoSyntaxSim each box picks its…	Its pin
2	Run Programme shows…	Each step as it runs
3	The sim writes 'Wait 3000ms'. On paper?	pause 3 s
4	How do you test a decision?	Press the input; watch the output
5	Name both simulators.	NoSyntaxSim and Yenka

Extension

EXTRA PRACTICE



1. Why is x3? better than 3 copies?

   Fewer boxes — and 3 copies scores 2 max

2. Where does the sub-procedure box go?

   Where a named set of steps is called

3. Two inputs both needed: how many decisions?

   Two, one after the other

Look back — by the end of this part…



1. I can build a flowchart in NoSyntaxSim and test it
2. I can build the same flowchart in Yenka
3. I can answer a past-paper flowchart question

LOOKING AHEAD

Lesson 3: Reading Programs

The exam also writes it as lines of code. Can you read them?

Course Specification

Digital electronic control systems

- using correct symbols (start, stop, input, output, branch and loop) to construct flowcharts showing solutions to simple control programs, involving time delays and continuous and fixed loops

Credits (1 of 3)

The big question (The Uno we use): Photo: JrawX, CC0, via Wikimedia Commons

The Six Symbols (data booklet p.6): © Qualifications Scotland (SQA). N4/N5 Engineering Science Data Booklet, p.6, Symbols for Flowcharts.

PRACTICE: Programming Structures (booklet page): Booklet 4b pages by R Stewart.

How the Marks Are Given (marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2021 Q13(b), 2022 Q9(a), 2023 Q8(a), 2024 Q11(b).

Worked Example: The Answer (marking-instruction layout): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2021 Q13(b), 2022 Q9(a), 2023 Q8(a), 2024 Q11(b).

Credits (2 of 3)

Spot the Mistake (marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2021 Q13(b), 2022 Q9(a), 2023 Q8(a), 2024 Q11(b).

PRACTICE: Flowcharts (booklet page): Booklet 4b pages by R Stewart.

NoSyntaxSim: Flowchart to Uno: NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

Build It: 2022 Q9 (question paper): © Qualifications Scotland (SQA). N5 Engineering Science 2022 Q9 and 2024 Q11, question papers and marking instructions.

Past Papers: Flowcharts (2024 Q11): © Qualifications Scotland (SQA). N5 Engineering Science 2022 Q9 and 2024 Q11, question papers and marking instructions.

Credits (3 of 3)

2022 Q9(a): 10 Marks, 1 per Line (marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science 2022 Q9 and 2024 Q11, question papers and marking instructions.

2024 Q11(b): 11 Marks, 1 per Line (marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science 2022 Q9 and 2024 Q11, question papers and marking instructions.

Course Specification: © Qualifications Scotland (SQA). Course spec v2.0, p.5

R Stewart - October 2026